Using LMs to Model the Effects of Context and Coreference during Sentence Comprehension

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Abstract

Language models (LMs) are often used as a tool to model human language processing. Recent studies suggest that severely restricting LMs' context window improves their fit to human psycholinguistic data by simulating human working memory constraints. However, it is possible that this strict memory-decay approach overlooks humans' reliance on long-range structural representations, such as discourse structre. In this work, we systematically vary the context window size of GPT-2 across four large-scale naturalistic English reading-time datasets and observe a U-shaped relationship: Although restricted contexts ($<$ 20 tokens) successfully capture local memory limitations, expanded contexts (500–1,000 tokens) ultimately yield the highest overall psycholinguistic fit. To investigate the mechanism driving this benefit, we conduct a *counterfactual* inference-time experiment that disrupts cross-sentential entity chains by pronominalizing repeated discourse entities. Obscuring these structural linkages significantly degrades the predictive power of larger context windows by 20% to 40%. Our experiments demonstrate that tracking long-range coreference relations is one important factor for the alignment between LM surprisal and human reading behavior, and approximate the extent to which human comprehenders use global discourse relations during language processing.[1]

1 Introduction

How do people integrate past information to predict what they will encounter next when reading or listening? In recent years, language models (LMs) have emerged not merely as engineering achievements (Radford et al., 2019; Brown et al., 2020), but as useful cognitive models to investigate human psycholinguistic mechanisms (Goldstein et al., 2022; Frank and Goodman, 2026). LMs' surprisal values have been demonstrated to be well correlated to human reading times (Wilcox et al., 2020; Oh et al., 2022; Kuribayashi et al., 2024, 2025). However, recent scaling of LMs has also revealed a noteworthy divergence: as models grow larger and achieve lower perplexity, their surprisal estimates often become less predictive of human reading behavior (Oh and Schuler, 2023; Shain et al., 2024). One leading hypothesis suggests this discrepancy arises because modern LMs are effectively "superhuman" (Oh and Linzen, 2026) in their prediction capabilities, possessing a long tail of factual knowledge and lossless short-term memory that bypass the cognitive constraints inherent to human language processing.

To better align model predictions with human reading behavior, recent computational investigations have attempted to restrict models' access to prior context, simulating human working memory limits (Hahn et al., 2022; Kuribayashi et al., 2022; Timkey and Linzen, 2023; De Varda and Marelli, 2024; Clark et al., 2025; Xu et al., 2026). Studies have shown that constraining the context window size (Kuribayashi et al., 2022) or introducing a linear recency bias to the attention mechanism (De Varda and Marelli, 2024; Clark et al., 2025) significantly improves the fit of LM surprisal to naturalistic human reading times. Yet, this strict memory-decay approach introduces a theoretical gap. It overlooks how people actively build and maintain structured discourse representations over extended narratives (Karttunen, 1969; Kintsch and van Dijk, 1978; Grosz et al., 1995; Ericsson and Kintsch, 1995; Kintsch, 1998; Jaffe et al., 2018; Tsipidi et al., 2024). For instance, reading times have been shown to be highly sensitive to long-range contextual information, such as the number of possible coreferences (Jaffe et al., 2018). Furthermore, the distribution of information across a text is not merely random, but rather fluctuates systematically

[1]Our code is available at https://github.com/kohei-kaji/contextsize.

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according to hierarchical discourse structures (Tsipidi et al., 2024, 2025). This suggests that comprehenders actively use long-range discourse linkages to drive forward-looking predictions. Such capacities have been treated theoretically, for example, by the *long-term working memory* framework (Ericsson and Kintsch, 1995; Kintsch, 1998), which posits skilled cognitive mechanisms for maintaining extended narrative states. Our hypothesis is that models that cannot take advantage of such discourse linkages will be insufficient as models of human language processing.

We investigate how context length and discourse structure influence the cognitive plausibility of language models: we systematically vary the context window size of GPT-2 and evaluate its predictive power for human reading time in English. We observe a U-shaped relationship between context size and psychometric predictive power. At short timescales ($<$ 20 tokens), restricted context provides a better fit to human data compared to longer one, successfully capturing localized working memory constraints observed by Kuribayashi et al. (2022). Crucially, however, at larger time scales ($>$ 20 tokens), expanded contexts become highly beneficial once again, with predictive power peaking at extended lengths of roughly 500 to 1,000 tokens. This U-shaped curve suggests that reading times reflect the interplay of two factors: local working memory constraints and global discourse integration.

Second, we ask *why* this extended context facilitates the predictive power of surprisal. We hypothesize that larger context windows are beneficial not merely because they provide a larger *bag of words*, but because coreference relations beyond sentence boundaries help maintain discourse coherence and enable people to predict what is coming next (Hobbs, 1979). While it is known that readers with greater working memory capacity are more successful at resolving pronouns (Daneman and Carpenter, 1980), the predictive benefits of an extended context window extend beyond simple pronominal anaphora. For example, consider the sentence pair: *Mary pushed Sue. The poor girl* ... Most comprehenders will establish a coreference link between *the poor girl* and *Sue*, and doing so dramatically reduces the entropy over possible next-word continuations. Similarly, Transformer-based LMs have been shown to establish coreference relations between noun phrases with or without lexical overlap (Tenney et al., 2019; Clark et al., 2019; Sorodoc et al., 2020). Therefore, we posit that LMs' ability to represent coreference cues partially accounts for their better alignment with human reading times as context size increases.

To test this, we examine LMs' predictions for reading times on *counterfactual* versions of the test materials where coreference relations are obscured. We analyze to what extent this degrades the models' ability to predict the reading times of people who have full knowledge of entity relationships. We find that when entity coreferences are pronominalized, the advantage in predictive power of larger context windows drops by approximately 20% to 40%, substantially above a baseline manipulation. This drop also provides a quantitative measure of the importance of coreference resolution in discourse processing. More broadly, our approach of reading-time modeling with a counterfactual context provides a generalizable framework for evaluating how specific aspects of context drive expectation-based human language comprehension.

2 Evaluating Surprisal on Reading Times

We adopt surprisal theory (Hale, 2001; Levy, 2008; Smith and Levy, 2013) as a framework for linking LMs' outputs and human psycholinguistic processing measures. Building on this framework, we investigate the relationship between LMs' context size and discourse structure.

2.1 Reading-time Datasets

We use four large-scale English reading-time datasets: Brown (Smith and Levy, 2013), Natural Stories (Futrell et al., 2021), OneStop (Berzak et al., 2025), and Provo (Luke and Christianson, 2018).[2] Brown consists of 13 passages (7,234 words) of **self-paced reading** (SPR) data from 35 native English speakers. Natural Stories consists of 10 naturalistic narratives (10,256 words). We use SPR data from 181 native English speakers (Futrell et al., 2021) and **A-Maze** reading times from 95 native English speakers (Boyce and Levy, 2023). OneStop consists of 10 articles (35,181 words) with eye-tracking data from 180 native English speakers. We use the "ordinary reading" sub-portion. Provo consists of 55 paragraphs (2,745 words) of eye-tracking data from 84 native English speakers. Table 1 summarizes the descriptive statistics of document length in GPT-2 tokens for each dataset.

[2] Although Kuribayashi et al. (2022) evaluated their models on the Dundee corpus (Kennedy et al., 2003), we exclude this dataset from our study because it features only 10 participants and is not publicly available.

Dataset	Mean	SD	Range
Brown	707.00	199.59	313–959
Natural Stories	1237.30	68.28	1156–1359
OneStop	751.27	171.57	450–1271
Provo	59.53	8.20	43–80

Table 1: Descriptive statistics (mean, standard deviation, and range) of the number of tokens by the GPT-2 tokenizer per document across datasets. Provo is notably shorter than the others.

In SPR, participants read texts word-by-word by pressing a button, with the interval between presses recorded as the reading time. In A-Maze, a variant of the Maze task (Forster et al., 2009), participants sequentially choose the correct next word from two options, and this choice time is taken as the reading time. For the eye-tracking datasets, we analyze three eye-tracking measures: **first fixation** (FF), the duration of initial fixation to a region during the first pass; **gaze duration** (GD), which sums all fixations on a region before the reader's gaze leaves the region in any direction; and **total fixation** times (TF), the sum of all fixations on a region.

We apply several preprocessing steps. For all datasets, we exclude the first and last words of each sentence and any words containing punctuation. For SPR data (Brown and Natural Stories), we follow Futrell et al. (2021) and exclude reading times outside the 100–3,000 ms range. We also exclude data from participants with poor comprehension, specifically those scoring less than $5/6$ on the comprehension questions for NS SPR and under 80% accuracy on A-Maze (Boyce and Levy, 2023). For eye-tracking data, we remove skipped words. For all datasets, we model the mean processing time across participants (Smith and Levy, 2013; Wilcox et al., 2023) to isolate item-level variance and ensure computational feasibility, though we acknowledge the inability to model subject-level random effects as a limitation.

2.2 Surprisal Estimates

We use four pretrained GPT-2 models (`Small`, `Medium`, `Large`, and `XL`),[3] as prior work has shown that GPT-2 surprisal, particularly by the `Small` variant, is the best predictor of English reading times compared to larger language models (Oh and Schuler, 2023; Shain et al., 2024). To estimate surprisal, we adopt the whitespace-trailing decoding (Oh and Schuler, 2024), which reassigns the probability of a leading whitespace to the preceding word. While GPT-2 utilizes at most 1,024 tokens to predict the next token, the use of this decoding strategy restricts our maximum context window to 1,023 tokens. We use 28 different context lengths (in subword tokens) from 1 to 1,023 to estimate surprisal for each word.[4] Following Kuribayashi et al. (2022), the BOS token is not included in the context as it can cause LMs to misinterpret the beginning of the sentence. The mean surprisal values for each dataset across all context sizes are provided in Figure 5 in Appendix B.

2.3 Regression Analysis

To quantify the predictive power of LM-based surprisal on reading time, we follow the methodology established in prior psycholinguistic studies (e.g., Frank and Bod, 2011; Goodkind and Bicknell, 2018; Wilcox et al., 2020, 2023) and measure the improvement in regression model fit afforded by surprisal (Δ_{LL}). Specifically, we compare the log-likelihood of a baseline regression model (excluding surprisal) against that of a target regression model (including surprisal) on held-out test data.

More precisely, the dataset $\mathcal{D} = \{(y_t, \mathbf{x}_t)\}_{t=1}^{N}$ is defined such that y_t represents the mean reading time for word w_t, $\mathbf{x}_t$ is a vector of predictors for w_t, and N is the total number of data points. We fit a linear regression model of the following form:

$$y_t = \beta_0 + \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \qquad (1)$$

where β_0 is the intercept and $\boldsymbol{\beta}$ is the vector of coefficients. The predictors $\mathbf{x}_t$ in the baseline model include word position within a document, word length (number of characters), and unigram surprisal estimated from about 33 billion tokens of the Pile dataset (Gao et al., 2020). We also incorporate two spillover positions (i.e., the values at $t-1$ and $t-2$) for both word length and unigram surprisal. The target model is constructed by adding surprisal of the current word, along with its two spillover positions, to the baseline model. All predictors are z-scored.

The predictive power of surprisal is evaluated with 10-fold cross-validation. The model parameters are estimated on the nine folds and evaluated on the held-out fold. For every data point t in the held-out fold, we compute the difference in log-

[3] We use the Transformers library (Wolf et al., 2020)

[4] The context sizes adopted in this study: [1, 2, 3, 4, 5, 6, 7, 10, 12, 16, 20, 25, 32, 40, 50, 64, 80, 100, 128, 160, 200, 256, 320, 400, 512, 640, 800, 1023].

likelihood between the target and baseline models:

$$\Delta_{LL}^{(t)} = \ell_{\text{target}}^{(t)} - \ell_{\text{baseline}}^{(t)}, \quad (2)$$

where $\ell_{\text{model}}^{(t)}$ is the log-likelihood for a single observation at t under the normal distribution defined in Equation (1). The final Δ_{LL} metric is calculated as the mean $\Delta_{LL}^{(t)}$ across data points in $\mathcal{D}$.

Thus, Δ_{LL} quantifies the extent to which LM surprisal accounts for reading-time variance beyond what is already explained by the baseline model.

3 Experiment 1: The Effect of Context Window Size

To investigate how an LM's context size influences the alignment between its surprisal and human reading times, we systematically vary the number of context tokens provided to GPT-2 models (Section 2.2). We then evaluate the predictive power of the resulting surprisal by using the method in Section 2.3 across the four large-scale English reading-time datasets described in Section 2.1.[5]

3.1 Results

The relationship between context size and the fit of surprisal to reading times is illustrated in Figure 1. Looking at the macro trend, except for Provo FF, we observe that expanding the context window size enhances predictive power across datasets and model variants. This trend is robust in datasets featuring longer documents (Brown, Natural Stories, and OneStop), whereas it is less consistent in Provo, which consists of shorter documents (see also Table 1). To test this trend empirically, we fit a Bayesian mixed-effects model using `brms` (Bürkner, 2021) in R (R Core Team, 2025) with Δ_{LL} as the response, log-transformed context length as a fixed effect, and model and dataset as random intercepts. We observed a robust positive effect of context length ($\hat{\beta} = 4.53 \times 10^{-3}$, 95% CrI: $[4.23, 4.83] \times 10^{-3}$).[6]

However, a closer inspection of shorter context windows (up to roughly 20 tokens) reveals a more nuanced pattern. With the exception of Natural Stories A-Maze, we observe a U-, or sometimes S-shaped pattern. This U-shape is especially pronounced in larger models and in measures that reflect earlier stages of cognitive processing, such as first fixation (FF) in eye-tracking. Within this short-context regime, the *longer-is-better* relationship does not strictly hold. Instead, local optima emerge at context sizes of approximately 2 to 4 tokens, though the exact peak varies depending on the model scale and dataset. This finding is consistent with Kuribayashi et al. (2022), who demonstrated that restricting context to within sentence boundaries can yield a better fit to human reading data.

These results suggest a dual mechanism of context effects: within the short-context region, a *shorter-is-better* trend operates locally, indicating that reading times are highly sensitive to immediate, local statistics. Conversely, as the model is granted access to increasingly longer cross-sentential context, the *longer-is-better* trend emerges, ultimately yielding the highest overall fit.

Finally, although tangential to our primary investigation of context effects, we observed a notable result regarding model scale on OneStop, a dataset released in 2025 with the largest number of participants (Berzak et al., 2025): surprisal from GPT2-`Medium` consistently outperforms that from GPT2-`Small` as a predictor of reading times. This deviates from the prevailing consensus in recent literature, which has identified surprisal from GPT2-`Small` as the best predictor of English reading times (Oh and Schuler, 2023; Shain et al., 2024).

3.2 Follow-up Analysis: Localization of the Context Effect

While Experiment 1 demonstrated that extended contexts yield the highest overall fit, the underlying mechanism remains unclear. To investigate, we analyze the relationship between part-of-speech (POS) tags and word-by-word variations in both raw surprisal and model fit. We hypothesize that if long contexts are useful specifically because they help to accurately model discourse state, then the predictive gains should fall on words that are important for establishing discourse relations (e.g., coreference relations and predicate-argument structures; Hobbs, 1979; Grosz et al., 1995), such as nouns and predicates, but not on determiners and particles.

We use Stanza (Qi et al., 2020) to assign Universal Dependencies POS tags (de Marneffe et al., 2021) to each word. To ensure statistical relia-

[5] As one of the reviewers suggested, we also evaluated models other than GPT-2 on the Natural Stories SPR. The results, presented in Appendix C, show that our overall findings remain consistent across different models.

[6] We assigned weakly informative priors to the fixed effect ($\mathcal{N}(0, 0.1)$) and variance components ($\text{Exponential}(1)$). The model was estimated using 4 Markov chains with 2,000 iterations per chain, including 1,000 warmup iterations. All parameters successfully converged ($\hat{R} < 1.01$).

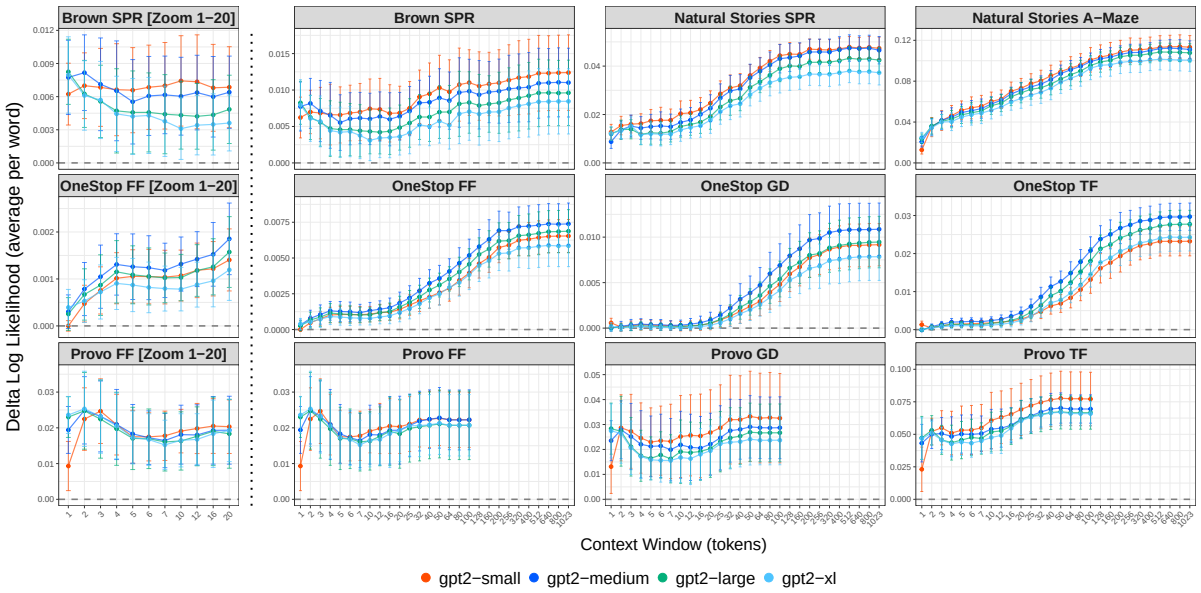


Figure 1: Predictive power (Δ_{LL}) across different context window sizes. Each point represents the Δ_{LL} with 95% confidence interval. For the Provo dataset, results are shown up to the document's maximum length. The leftmost column displays zoomed-in plots for Brown, OneStop FF, and Provo FF, focusing on context windows of 1–20 tokens. Abbreviations: SPR = self-paced reading; FF = first fixation; GD = gaze duration; TF = total fixation time.

bility, our analysis is restricted to major syntactic categories with at least 100 occurrences across the datasets. The evaluated tags include open class words (ADJ, ADV, NOUN, PROPN, VERB) and closed class words (ADP, AUX, CCONJ, DET, NUM, PART, PRON, SCONJ).

To isolate the localized advantage of long-range context, we compare the Small estimates under a *short* context baseline (2 tokens) against a *long* context setup (up to 1,023 tokens or the maximum length permitted by the document). For each word position t, we evaluate two metrics. First, to localize and quantify the shifts in surprisal values induced by an expanded context window, we define the delta surprisal as:

$$\Delta\,\mathrm{surp}^{(t)} = \mathrm{surp}_{\mathrm{Short}}(t) - \mathrm{surp}_{\mathrm{Long}}(t). \quad (3)$$

Second, to capture the incremental improvement in predictive power for reading time, we compute the pointwise change in log-likelihood between the long and short context regression models:

$$\Delta\Delta_{LL}^{(t)} = \ell_{\mathrm{Long}}^{(t)} - \ell_{\mathrm{Short}}^{(t)}, \quad (4)$$

where $\ell_{\mathrm{context}}^{(t)}$ denotes the log-likelihood at t under the respective context size condition.

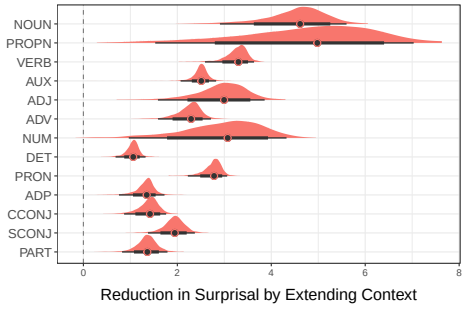
To assess these localized effects while accounting for dataset-specific variance, we fit Bayesian mixed-effects models with brms. We model both $\Delta\,\mathrm{surp}^{(t)}$ and $\Delta\Delta_{LL}^{(t)}$ as functions of the categorical POS predictors, and include random slopes for the respective datasets.

3.2.1 Results

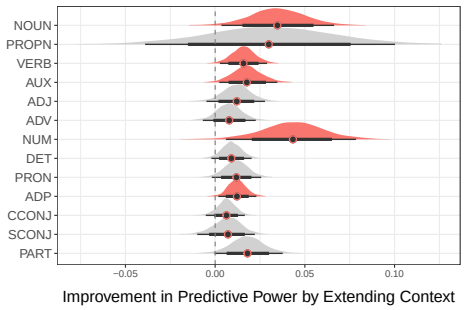
The posterior estimates reveal a divergence between the reduction in model surprisal and the improvements in reading-time prediction.[7] As shown in Figure 2a, expanding the context window size systematically reduces surprisal across all evaluated POS tags. This surprisal reduction is most pronounced for proper nouns (PROPN), nouns (NOUN), verbs (VERB), numerals (NUM), and adjectives (ADJ).

However, these localized drops in surprisal do not uniformly translate to better psychometric fit (Figure 2b). Most notably, while PROPN exhibits the largest surprisal reduction, its improvement in predictive power ($\Delta\Delta_{LL}^{(t)}$) is not statistically significant due to wide variance across datasets. Instead, the statistically significant gains in $\Delta\Delta_{LL}^{(t)}$ are concentrated in NOUN, NUM, VERB, auxiliry (AUX), and adpositions (ADP). That being said, the two biggest gains from larger contexts are assigned to words establishing discourse relations such as nominals and predicates, while determiners and particles, for

[7] We assigned weakly informative priors to the fixed effects ($\mathcal{N}(0,2)$ for $\Delta\,\mathrm{surp}^{(t)}$ and $\mathcal{N}(0,0.1)$ for $\Delta\Delta_{LL}^{(t)}$) and variance components ($\mathrm{Exponential}(1)$). Both models were estimated using 4 Markov chains with 2,000 iterations per chain, including 1,000 warmup iterations. All parameters successfully converged ($\hat{R} < 1.01$).



(a) Posterior estimates of $\Delta\,\mathrm{surp}^{(t)}$ across POS tags. Positive values indicate that long-range context systematically reduces the model's surprisal for that category.



(b) Posterior estimates of $\Delta\Delta_{LL}^{(t)}$ across POS tags. Positive values indicate a greater reading-time predictive benefit from an extended context window at that category.

Figure 2: Localization analysis of context effects across POS tags using Bayesian mixed-effects models. Densities show the central 99% of the posterior distributions. Points represent posterior medians, while the thick and thin horizontal bars denote the 80% and 95% credible intervals, respectively. Red distributions indicate that the 95% credible interval does not contain zero.

example, show no significant gains from context, which is consistent with our original hypothesis.

These findings demonstrate that an extended context window does not merely act as a uniform mechanism for benefiting prediction. Rather, the predictive gains are concentrated primarily on nominals and predicates. We hypothesize that these categories benefit most because they serve to represent discourse entities and events. This suggests that expanding the context window enables the model to implicitly track coreference relations and maintain the global discourse states. We turn to this question in the next section.

4 Experiment 2: Disrupting Coreference Relations during Inference

An important aspect of language processing is figuring out who is doing what to whom. This type of information is supplied by referring expressions that provide links to discourse entities.

During comprehension, people incrementally update their mental model about the entities being referred to by tracking the coreference relations (Karttunen, 1969). Coreference resolution, the task of establishing the identity between two referring expressions with or without the same wording, is also an important NLP task (Jurafsky and Martin, 2026). LMs have also been shown to establish coreference relations across long documents (Tenney et al., 2019; Clark et al., 2019; Sorodoc et al., 2020). We hypothesize that it is specifically this ability that drives the alignment between long-context surprisal and human reading time. We predict that when a model's ability to track entities is hindered, the advantage of long-context prediction will be degraded.

In order to test this prediction, we perform inference-time manipulations in which we pronominalize all entities that occur more than once, creating a COREF-DISRUPTED version of the test materials used in Experiment 1. This removes information on the entity required to establish unambiguous coreference links with other mentions of the same entity in the context. We then analyze to what extent this degrades the model's ability to predict reading times of people who have access to the full, discourse-linked context. Decrease in reading time prediction indicates that the counterfactual context loses the structural information useful for simulating human language processing.

However, the pronominalization manipulation will significantly alter the text—for example, changing its overall entropy—and may lead to unnatural fluctuations in surprisal. To account for these confounds, we create a SINGLETON-DISRUPTED version of the original text where we pronominalize all the entities that appear only once. This version of the text is pronominalized, but long-context dependency information is preserved; thus, reading time prediction should degrade only in the COREF-DISRUPTED condition (not in SINGLETON-DISRUPTED) if the coreference chain is important for simulating human reading. Overall, 21,640 tokens are changed in SINGLETON-DISRUPTED and 22,695 tokens are changed in COREF-DISRUPTED. The number of tokens changed in SINGLETON-DISRUPTED and COREF-DISRUPTED are roughly equivalent, ensuring that the changes in delta log-likelihood do not just result from pronominalization but from the loss of long-context dependencies.

To illustrate our two conditions, consider the

examples below adapted from Brown, where (1) is SINGLETON-DISRUPTED and (2) is COREF-DISRUPTED. Pronominalized text is colored. Version (1) contains several local infelicities, but maintains information about the main entities of the narrative. Version (2) is perfectly fluent, but makes all references to entities ambiguous.

(1) Customers hastily vacated their tables as the tall buffalo hunter pushed open them and walked towards it. Only Blue Throat stayed where he was, his supporting his. He leered at the stranger.

(2) They hastily vacated their tables as he pushed open the doors and walked towards the bar. Only he stayed where he was, his elbows supporting his massive frame. He leered at him.

Both texts are out of distribution for an LM trained on naturally occurring English. We predict that, even though texts like (2) may be perceived as more fluent, conditioning on them will yield surprisal estimates with lower predictive power for reading times, especially when the context window is long.

4.1 Methods

4.1.1 Dataset Creation.

Our methods are largely the same as those described in Section 2, except for the datasets on which surprisal values are conditioned. To create these datasets, we use the coreference parser from Stanza (Qi et al., 2020; Liu et al., 2024)[8] to tag entities in our test materials. We use silver entity tags to identify singleton and coreferent entities, which are replaced in the SINGLETON-DISRUPTED and COREF-DISRUPTED conditions, respectively. We use the following methods to replace entities: All tokens associated with an entity are replaced by a `[mask]` token. We do not mask out entities if part of the entity falls outside of a model's context window. We then prompt DeepSeek-V4-Flash (DeepSeek-AI, 2026) to collapse each contiguous `[mask]` span into a single appropriate pronoun, providing the original sentence and the masked sentence as context. If no pronoun is produced, we use *it* as a fallback. For the full prompt template, see Appendix E. We do not change entities that are already pronominalized.

This procedure results in three parallel texts. However, the SINGLETON-DISRUPTED and COREF-DISRUPTED conditions are shorter, as multi-token entities are replaced with a single-token pronoun. To account for this, we fill extra tokens from the previous context chunk in the same condition.

[8]The accuracy of this model is reported to be 95.7% on the CorefUD dataset (Nedoluzhko et al., 2022).

4.1.2 Text Comprehensibility Check

To validate the naturalness of these two versions after perturbations, we evaluate their comprehensibility. We conduct this evaluation on Natural Stories by having DeepSeek-V4-Flash answer reading comprehension questions based on the texts. To avoid the risk of data contamination, where the LLM might have memorized the publicly available questions from the original datasets, we create a customized set of comprehension questions for Natural Stories inputs. The questions are given in Appendix F. Overall, DeepSeek-V4-Flash achieves 100% (60/60) correct on the original text, 91.7% (55/60) on COREF-DISRUPTED and 90% (54/60) correct on SINGLETON-DISRUPTED.

4.1.3 Surprisal Estimation

We use the same GPT-2 models as in Experiment 1 to obtain surprisal values. We then calculate Δ_{LL} for each condition (SINGLETON-DISRUPTED and COREF-DISRUPTED) and subtract each from the Δ_{LL} value obtained for the unaltered text in Experiment 1. This metric is the same $\Delta\Delta_{LL}$ as in Section 3.2, but calculated against the unaltered text for each condition, as opposed to short vs. long contexts in Section 3.2. When computing surprisal for entities as target tokens, we always unpronominalize the entity first; the only difference between the conditions is therefore the context in which the surprisal is conditioned.

4.2 Results

Results for `Small` are shown in Figure 3, with $\Delta\Delta_{LL}$ plotted on the y-axis. Results for all models are presented in Appendix D, and are consistent across all models tested.

We observe two major trends. First, when conditioning on local contexts ($\lessapprox$ 10 tokens), $\Delta\Delta_{LL}$ is very close to zero, indicating little change from the unaltered baseline. In three cases (Brown SPR, Natural Stories SPR, and Provo TF), SINGLETON-DISRUPTED shows slightly negative $\Delta\Delta_{LL}$, likely due to the fact that SINGLETON-DISRUPTED disrupts local semantic predictability.

However, as the context size gets larger, COREF-DISRUPTED shows a large decrease $\Delta\Delta_{LL}$. This is greater than the decrease observed in the

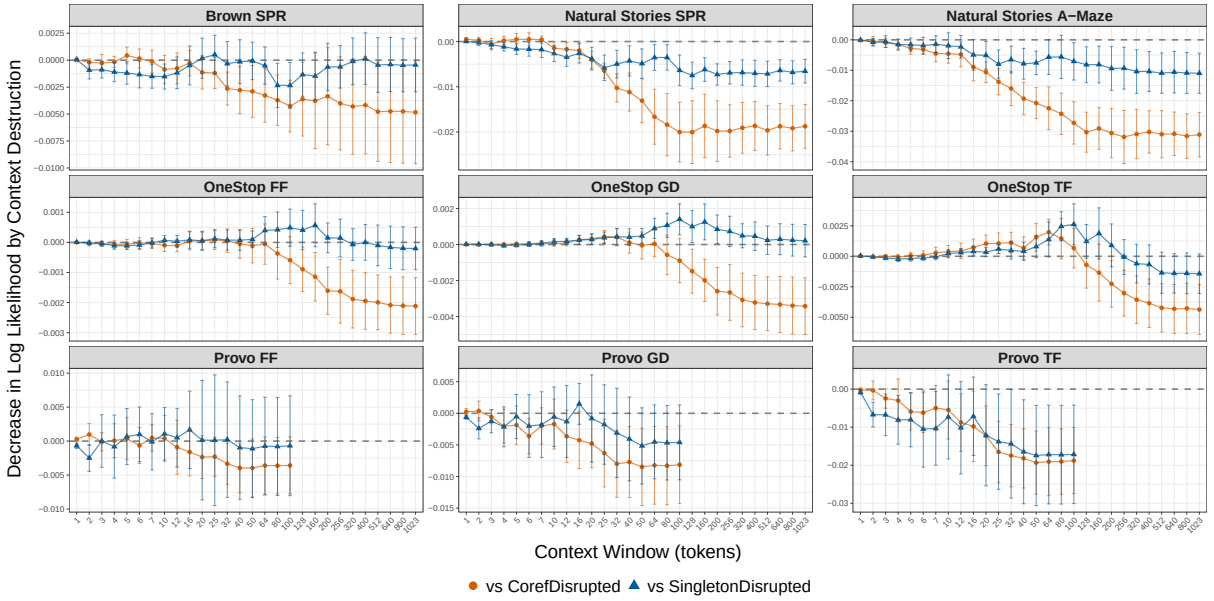


Figure 3: Difference of predictive power ($\Delta\Delta_{LL}$) between Experiment 1 and our two target conditions—SINGLETON-DISRUPTED and COREF-DISRUPTED. Results are for `Small`.

SINGLETON-DISRUPTED condition in most cases, except for Provo. To test this statistically, we conducted paired word-level two-tailed t-tests. At the maximal context length, the $\Delta\Delta_{LL}$ in the COREF-DISRUPTED condition is significantly lower than zero across all datasets at $p < 0.01$ except for Provo FF ($p = 0.091$), and significantly less than the $\Delta\Delta_{LL}$ in the SINGLETON-DISRUPTED condition for Natural Stories SPR, Natural Stories A-Maze, One Stop FF, and One Stop GD ($p < 0.001$ for all), as well as for Brown and One Stop TF ($p < 0.05$ for both). We do not observe differences between our two conditions in Provo, likely due to its short-length documents.

Figure 4 shows the overall percent reduction in Δ_{LL} at a maximum context resulting from our experimental manipulations. We find that obscuring coreference relations can reduce the predictive power of the LM by 16.5% (Provo FF) to 42.4% (OneStop GD).

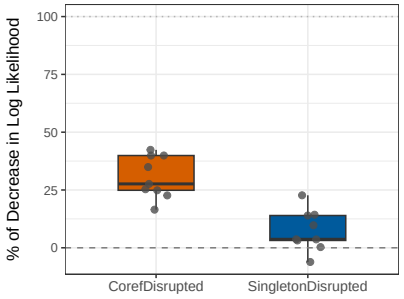


Figure 4: Distribution of the percentage decrease in log-likelihood under COREF-DISRUPTED and SINGLETON-DISRUPTED conditions for `Small` at a maximum context across all datasets.

5 Discussion

5.1 The Dual Mechanism of Context Length

The U-shaped predictive curve observed in Experiment 1 reveals the dual mechanism of context length for modeling human reading behavior. Our results indicate that human reading times may reflect two distinct processing constraints simultaneously: a *short-term* prediction mechanism that is influenced by working memory limits and a *long-term* prediction mechanism that is required to track long-range discourse structures. Recent computational psycholinguistic literature has predominantly focused on the former, demonstrating how severe context restrictions improve reading-time predictions (Hahn et al., 2022; Kuribayashi et al., 2022; Timkey and Linzen, 2023; De Varda and Marelli, 2024; Clark et al., 2025; Xu et al., 2026). The present study broadens this perspective by showing the latter's contribution: reading times for long documents are better predicted by surprisal from language models with longer context windows.

This dual-mechanism framework raises a fundamental question for future computational modeling: how is global discourse information compressed and represented within human working memory? Prior research has shown memory compression for

prediction at the level of individual tokens (Hahn et al., 2022) and syntactic structures (Kajikawa and Isono, 2026; Isono and Kajikawa, 2026). Extending these focuses to higher-level semantic representations, such as coreference chains and entity states, remains a critical open challenge. Investigating the compression and/or representation through the lens of language models offers a promising avenue for testing cognitive theories of working memory (Ericsson and Kintsch, 1995; Kintsch, 1998) that posit skilled mechanisms for processing extended narrative states.

It is noteworthy that the long-context effect was prominent in extended texts (e.g., Natural Stories, averaging 48.5 sentences per document and 21.1 words per sentence) but minimal in Provo, a shorter dataset (2.5 sentences and 13.3 words per sentence). This suggests that reading strategies may vary by document length. To extend this investigation across languages, multilingual reading-time datasets like MECO (Siegelman et al., 2022, 2025) can serve as a useful resource. However, as they primarily consist of short texts (e.g., an average of 8.3 sentences in English MECO), developing datasets with longer narratives in multiple languages remains critical to further explore these long-range effects.

5.2 Quantifying Discourse Contributions

Our findings reveal that the improvement of predictive power of extended context for reading time relies heavily on structured discourse relations. While psycholinguistic literature has long established that human readers use entity coreference to construct coherent representations during comprehension (Kintsch and van Dijk, 1978; Grosz et al., 1995; Jaffe et al., 2018), our study isolates and quantifies this process through computational modeling with LMs and surprisal theory as a linking method. By applying a pronominalization ablation strategy, we demonstrated that obscuring structural linkages during inference degrades the predictive power of larger context windows by approximately 20% to 40%. This substantial drop along with the relatively smaller decreases in our control conditions indicates that LMs do not merely treat extended context as an unstructured bag of words. Rather, they implicitly maintain and update entity states, mirroring human discourse comprehension.

Beyond coreference, our inference-time masking methodology offers a generalizable framework to decompose other aspects of contextual influence. By selectively disrupting specific structural linkages, researchers can apply this targeted ablation approach to quantify other factors that have been hypothesized to drive discourse representations, such as temporal event structures (Zwaan et al., 1995; Pustejovsky et al., 2003) or rhetorical relationships (Mann and Thompson, 1988; Zeldes et al., 2025). Future work should also investigate whether similar methods, applied at training time, yield similar results. We believe that methods like ours can help move away from treating context as a monolith within computational modeling frameworks. They have the potential to help identify and empirically validate the specific linguistic signals that drive behavioral alignment.

6 Conclusion

We demonstrated a U-shaped relationship between language model context size and psychometric predictive power across four English reading-time datasets. While restricted contexts successfully capture local working memory constraints, expanded windows (500–1,000 tokens) yield the best overall fit. Our inference-time pronominalization experiment reveals that this long-context advantage is significantly driven by entity chains, as obscuring cross-sentential coreference degrades predictive power by up to 42.4%. These results confirm that humans leverage global discourse relations during comprehension, demonstrating that accurate cognitive models must simultaneously capture both local working memory limits and long-range discourse integration.

Limitations

First, our empirical findings are based exclusively on English reading-time datasets. English relies heavily on explicit pronouns and rigid word order to establish coreference. In pro-drop languages or morphologically rich languages, coreference relations are often mediated by verbal morphology and/or zero anaphora. The precise nature of the U-shaped context effect and the relative contribution of entity chains to surprisal fit may vary cross-linguistically. Future research should extend this counterfactual methodology to typologically diverse languages to verify the consistency of these dual mechanisms.

Second, while our intervention finds that LMs leverage coreference signals to improve predictability, it does not confirm that LMs and humans con-

struct the same kind of structured representations.

Third, our counterfactual language modeling paradigm obscures coreference relations solely at inference time. As we manipulate the input texts, the resulting evaluation data becomes slightly out-of-distribution relative to the models' naturalistic training corpora. The models are therefore evaluated on coreference patterns they did not naturally observe during pre-training. Future work could extend our paradigm to counterfactual training settings, where models are trained from scratch on corpora with systematically altered coreference structures to better isolate these processing effects.

Finally, our experimental pipeline relies on silver-standard annotations to identify and manipulate coreference relations. Errors in the underlying automated coreference resolution could occasionally lead to imperfect counterfactual replacements. Future studies could validate these findings on gold-standard human-annotated corpora to ensure that errors in automated annotation do not systematically bias the evaluation of context effects.

Ethical Considerations

This study relies on the computational analysis of publicly available, anonymized human reading-time datasets and open-source language models. No new data from human participants was collected for this work. To the best of our knowledge, all utilized artifacts were originally collected and released in accordance with standard ethical guidelines.

AI Writing/Coding Assistance Policy

We used generative AI tools solely for the purpose of adjusting the grammar and phrasing of the manuscript, and a coding assistant for formatting tables and figures.

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A Artifacts

A.1 Language Models

Table 2 lists the language models used in Sections 3 and 4.

A.2 Data and Tools

Table 3 lists the datasets and tools used in Sections 3 and 4. We used these artifacts according to their licenses. For computing surprisal, we used a single NVIDIA H100 GPU (80GB) for several hours.

B Mean Surprisal

Figure 5 shows the mean surprisal values for each dataset across all context sizes examined in this study.

C Experiment 1 Results on Other Model Families

In addition to the GPT-2 series in Experiment 1 (Section 3), we evaluate the predictive power of other language model families, specifically the GPT-Neo (Black et al., 2021), GPT-J (Wang and Komatsuzaki, 2021), and OPT (Zhang et al., 2022) series, on Natural Stories SPR to ensure that our findings generalize across different architectures.

Figure 6 shows the result. For models with a maximum context length of 2,048 tokens, we also evaluate and plot the Δ_{LL} at their maximum context length. Consistent with our main findings, we observe a U-, or sometimes S-shaped pattern regardless of the model family.

D Full Results for Experiment 2 across All Models

Figures 7 to 9 illustrate the results of Experiment 2 in Section 4 for GPT2-`Medium`, `Large`, and `XL`, respectively. Results are consistent with those reported in the main body of the text: The COREF-DISRUPTED condition results in a substantial drop of predictive power relative to the baseline. The SINGLETON-DISRUPTED condition results in no or a smaller reduction in predictive power. As with our main results, Provo is an exception, likely due to its shorter context lengths.

Prompt template used for batch pronominalization. Braced fields denote values populated at inference time.

```
System:
You are a linguistic expert. You will receive sentences with placeholders like
[[TARGET_N]]. Your task is to choose exactly one appropriate English pronoun for
each placeholder. Use sentence position to choose case: subject, object,
possessive, or reflexive. Refer to the provided Context only to determine number,
animacy, and likely gender. Never return the original Context text as a value.
Return ONLY a valid JSON object. It must have this shape:
{"results": [{"sentence_index": 0, "pronoun_mapping": {"[[TARGET_0]]": "it"}}]}.
Every placeholder in Context must appear exactly once in pronoun_mapping.

User:
Sentence {sentence_index}: {masked_sentence}
Context: [[TARGET_0]]: {original_noun_phrase_0},
[[TARGET_1]]: {original_noun_phrase_1}, ...

Sentence {sentence_index}: {masked_sentence}
Context: [[TARGET_0]]: {original_noun_phrase_0},
[[TARGET_1]]: {original_noun_phrase_1}, ...
```

E Prompt template for substituting an entity span with a pronoun

We used the following prompt template to obtain pronoun replacements for masked noun phrases. Each span of contiguous masked tokens is replaced with a placeholder of the form `[[TARGET_N]]`. Each input sentence contained one or more placeholders. The corresponding context specified the original noun phrase for each placeholder. The model was instructed to return a JSON object mapping every placeholder to exactly one English pronoun. See the box below.

F Comprehensibility of the Discourse-disrupted Texts

To prevent data contamination from the publicly available Natural Stories questions,[9] we prompted Gemini-3.6-Thinking to generate new comprehension questions in the same style. We then prompted DeepSeek-V4-Flash with three versions of the text: unmodified original, SINGLETON-DISRUPTED, and COREF-DISRUPTED, along with the corresponding questions. The task was framed as a forced choice between options (a) and (b). While the correct answer below is always (a), option order was randomized per prompt.

Story 1

1. What surrounds the valley where the city of Bradford is located?
   (a) moors as high as mountains
   (b) dense oak forests
2. At what time of day did the fearsome boar prefer to come out?
   (a) in the middle of the day
   (b) in the dead of night
3. Why did the first huntsman decide to cut out the tongue instead of bringing the boar's head?
   (a) the head was too heavy for him to carry

[9] https://github.com/languageMIT/naturalstories/blob/master/naturalstories_RTS/Comp_questions_finalized.txt

Model	URL	#Params
GPT2-small (`Small`)	https://huggingface.co/openai-community/gpt2	124M
GPT2-medium (`Medium`)	https://huggingface.co/openai-community/gpt2-medium	355M
GPT2-large (`Large`)	https://huggingface.co/openai-community/gpt2-large	774M
GPT2-xl (`XL`)	https://huggingface.co/openai-community/gpt2-xl	1.5B
GPT-Neo 125M	https://huggingface.co/EleutherAI/gpt-neo-125m	125M
GPT-Neo 1.3B	https://huggingface.co/EleutherAI/gpt-neo-1.3B	1.3B
GPT-Neo 2.7B	https://huggingface.co/EleutherAI/gpt-neo-2.7B	2.7B
GPT-J 6B	https://huggingface.co/EleutherAI/gpt-j-6b	6B
OPT 125M	https://huggingface.co/facebook/opt-125m	125M
OPT 350M	https://huggingface.co/facebook/opt-350m	350M
OPT 1.3B	https://huggingface.co/facebook/opt-1.3b	1.3B
OPT 2.7B	https://huggingface.co/facebook/opt-2.7b	2.7B
OPT 6.7B	https://huggingface.co/facebook/opt-6.7b	6.7B
OPT 13B	https://huggingface.co/facebook/opt-13b	13B
OPT 30B	https://huggingface.co/facebook/opt-30b	30B
OPT 66B	https://huggingface.co/facebook/opt-66b	66B

Table 2: Pretrained language models used in this paper.

Artifact	License
Brown (Smith and Levy, 2013)	CC BY 3.0
Natural Stories (Futrell et al., 2021)	CC BY-NC-SA 4.0
OneStop (Berzak et al., 2025)	CC BY 4.0
Provo (Luke and Christianson, 2018)	CC BY 4.0
Transformers (Wolf et al., 2020)	Apache 2.0
Stanza (Qi et al., 2020; Liu et al., 2024)	Apache 2.0
brms (Bürkner, 2021)	GNU General Public License v2.0
DeepSeek-V4-Flash	MIT License
Gemini-3.6-Thinking	Proprietary

Table 3: Artifacts and their licenses used in this paper.

(b) he wanted to save the head for a feast

4. Where did the first huntsman put the boar's tongue after cutting it out?
   (a) in his hunting pouch
   (b) in his coat pocket
5. What defect did the Lord of the Manor immediately notice when inspecting the boar's head?
   (a) the tongue was missing
   (b) one of the tusks was broken
6. According to the ending of the story, whose fame was ultimately more lasting?
   (a) the fearsome Bradford Boar's
   (b) the victorious huntsman's

Story 2

1. What did Aqua wish to sail on when he grew tired of playing in the ocean?
   (a) a white cloud
   (b) a sea gull's wing
2. Into what form did the sun change the water drops so they were light enough to carry into the sky?
   (a) fine mist or vapor
   (b) soft snowflakes
3. Who gathered the frightened water drops into a heavy gray cloud to send them back down to earth?
   (a) the wind
   (b) the sun
4. What turned the tremendous wheel inside the flour mill?

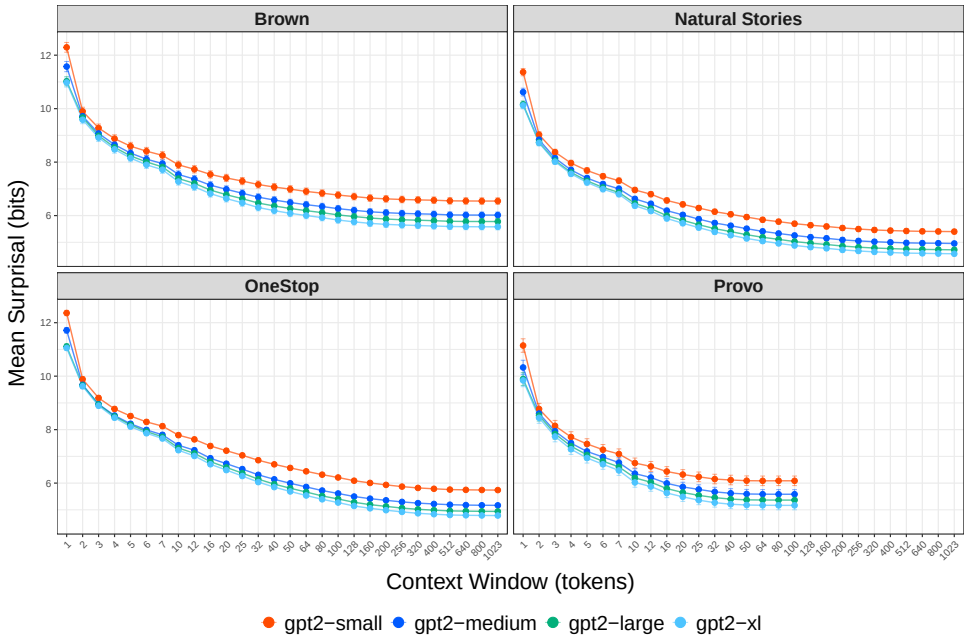


Figure 5: Mean surprisal as a function of context window size up to maximum length across four reading-time datasets. Each point represents the mean value and error bars show 95% confidence interval.

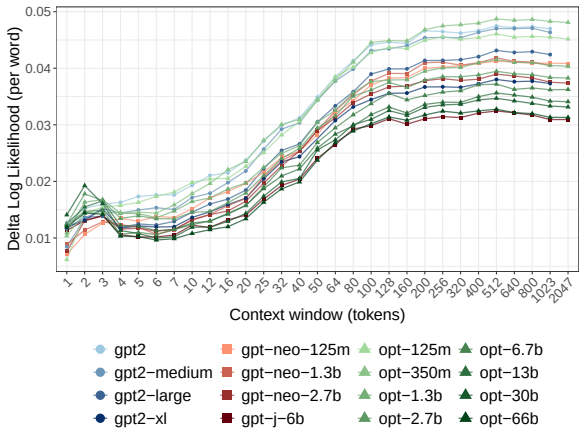


Figure 6: Predictive power (Δ_{LL}) across varying context window sizes for the GPT-2, GPT-Neo, GPT-J, and OPT model series on Natural Stories SPR.

(a) the tiny water drops
(b) a giant steam engine

5. What did Aqua use as horses while playing in the peaceful pond?
   (a) frogs
   (b) ducks
6. Where was Aqua at the very end of his journey?
   (a) back in the ocean
   (b) in a deep forest lake

Story 3

1. Why did the boy take one of the girl's lost slippers?
   (a) to use it as a home for his pet mouse
   (b) to sell it to a shoemaker
2. Where did the girl sit down to cover herself when she could not go home?
   (a) in a corner behind the little bakery
   (b) inside a wooden shed
3. What did the girl see when she lit her first match?
   (a) a large iron stove with brass feet
   (b) a grand banquet with roast goose
4. What did the girl do during her vision of the forest when the second match was lit?
   (a) dug up food hidden by a squirrel
   (b) picked a bouquet of white lilies
5. According to her grandmother, what happens when a star falls from the sky?

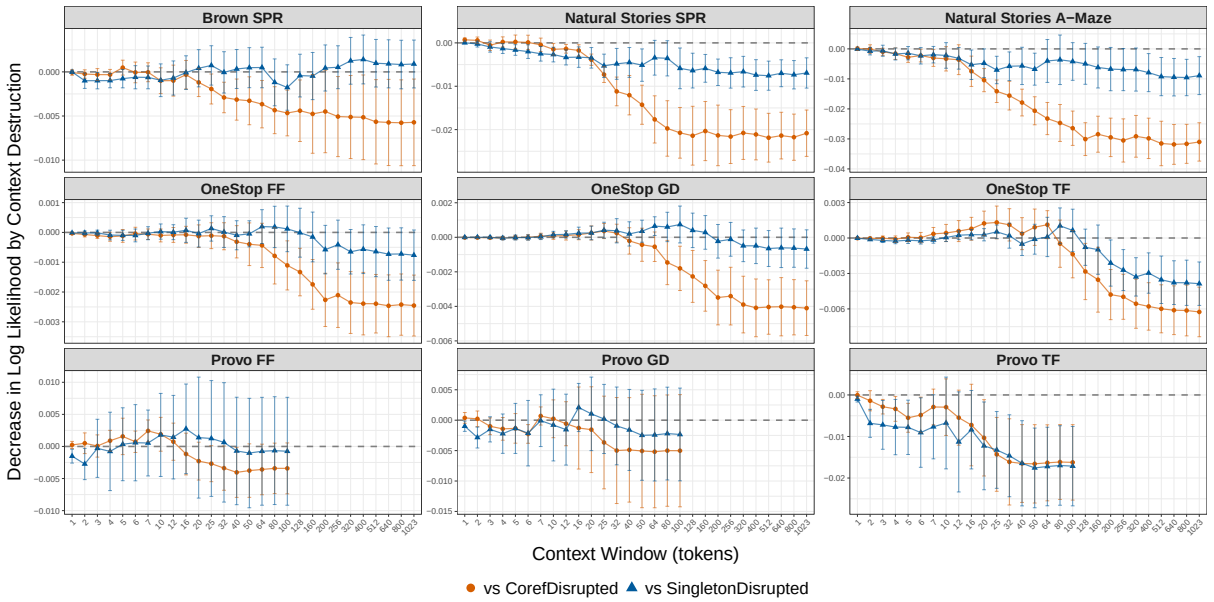


Figure 7: Difference of predictive power between the original version reported in Figure 1 and SINGLETON-DISRUPTED and between the original version and COREF-DISRUPTED across multiple context sizes in `Medium`.

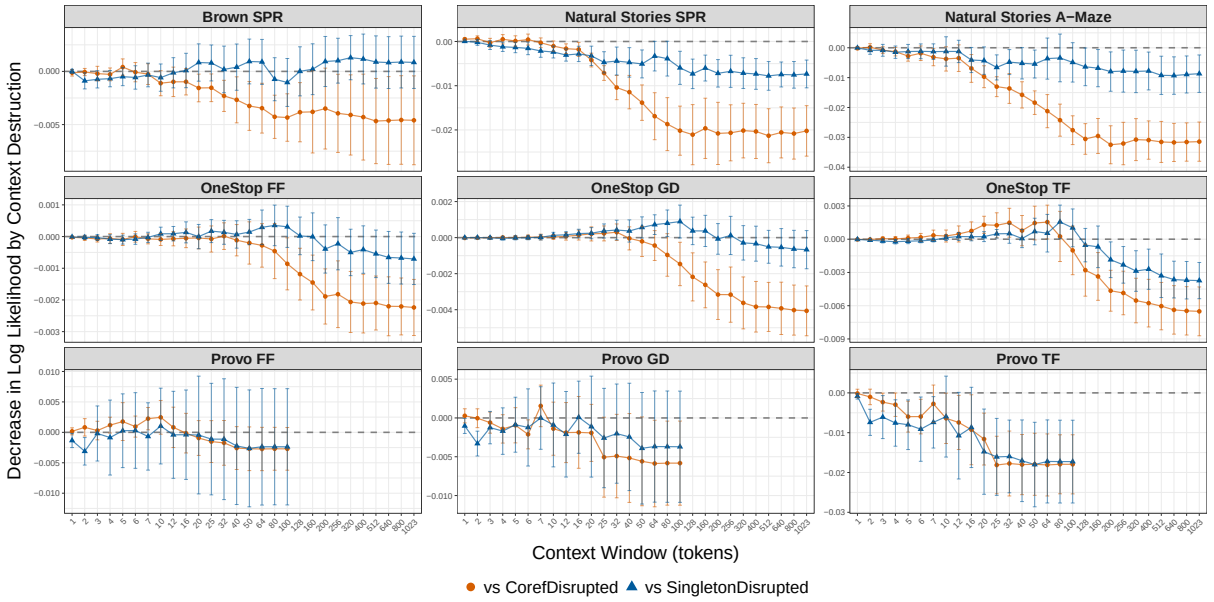


Figure 8: Difference of predictive power between the original version reported in Figure 1 and SINGLETON-DISRUPTED and between the original version and COREF-DISRUPTED across multiple context sizes in `Large`.

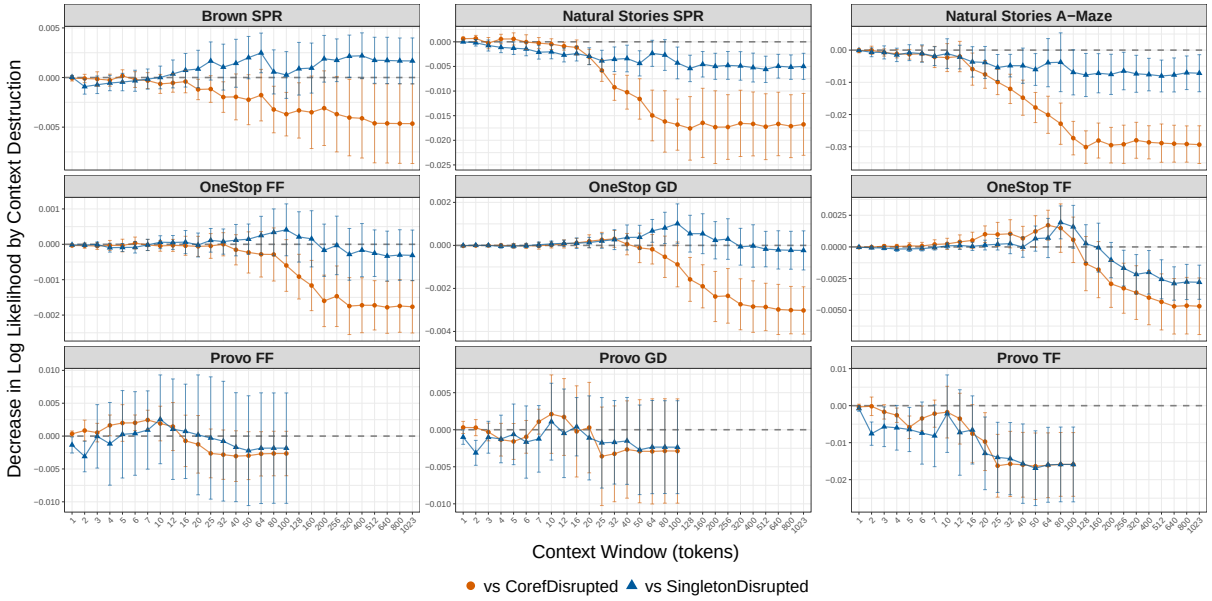


Figure 9: Difference of predictive power between the original version reported in Figure 1 and SINGLETON-DISRUPTED and between the original version and COREF-DISRUPTED across multiple context sizes in XL.

(a) a soul leaves this earth
(b) a child is born

6. What did the girl do to keep her grandmother from disappearing when the matches burned out?
   (a) lit the whole bundle of matches
   (b) sang a lullaby

Story 4

1. Where did the little nameless bird hide right before the flying competition began?
   (a) on the eagle's back
   (b) under the hawk's wing
2. Why did the other birds refuse to accept the little bird after he flew higher than the eagle?
   (a) he broke the rules by cheating
   (b) he was too weak to protect them
3. What was the second contest proposed to determine who would be king?
   (a) going the deepest into the earth
   (b) finding the largest piece of food
4. What hiding place did the little bird creep into for the second contest?
   (a) a pitch dark mouse's hole
   (b) a hollow oak tree
5. How did the little bird manage to escape from the owl?
   (a) the owl fell fast asleep with both eyes shut
   (b) the owl left to go hunting
6. What do the other birds mockingly call the nameless bird at the end of the story?
   (a) the hedge-king
   (b) the trickster-king

Story 5

1. Who explained to the narrator what made Elvis Presley so special?
   (a) Eugene Correthers
   (b) the orphanage matron
2. What city were the boys taken to for new shoes and haircuts?
   (a) Florida
   (b) Tennessee
3. What color did the bones in the boy's feet look through the x-ray machine at the shoe store?
   (a) green
   (b) blue
4. Who did the matron speak to after the young barber initially shook his head no?
   (a) a little man in a squeaky office chair
   (b) the shop owner waiting outside
5. What did the barber offer the boy after giving him a buzz cut?
   (a) a nickel for a candy bar
   (b) a quarter for a soda
6. What question did the boy ask the barber outside at the end of the story?
   (a) if Elvis Presley has green bones
   (b) if Elvis ever got a buzz cut

Story 6

1. What did Abby draw in art class using two pieces of expensive paper?
   (a) an elephant
   (b) a lion
2. What tool did Abby's mother use to clean out the fish tank?
   (a) a special vacuum
   (b) a water filter
3. What item did Abby retrieve to help search the tank for Mr. Sticky?
   (a) a magnifying glass
   (b) a flashlight
4. What did Abby briefly mistake for the water snail while searching?
   (a) a large speck of dust
   (b) a small pebble
5. Where in the fish tank was Mr. Sticky finally discovered hiding?
   (a) in a curve of the archway
   (b) inside a castle tower
6. What surprise addition did Abby and her mother find right next to Mr. Sticky?
   (a) another water snail
   (b) a plastic treasure chest

Story 7

1. How did Lucy respond when her sister asked if she had reported the text messages?
   (a) she lied and said she had
   (b) she admitted she was afraid to
2. Whose worn, oversized coat was Lucy wearing to protect herself from the rain?
   (a) her sister Jill's
   (b) her mother's
3. What animal was Lucy compared to in the very first abusive text message she received?
   (a) a giraffe
   (b) a flamingo
4. In what position was Lucy scheduled to perform in the talent show?
   (a) first
   (b) last
5. What encouraging gesture did Jill give Lucy from the wings right before the curtains opened?
   (a) a thumbs up
   (b) a wave
6. Who was cheering the loudest in the audience after Lucy finished singing her song?
   (a) her mother
   (b) her school teacher

Story 8

1. What classified program did the U.S. military attribute the recovered debris to?
   (a) Project Mogul
   (b) Project Echo
2. What did the Roswell Army Air Field's initial press release on July 9, 1947, claim personnel had recovered?
   (a) a crashed flying disc
   (b) a fallen weather balloon
3. Who did Stanton Friedman interview in 1978 about transporting the debris to Fort Worth?
   (a) Jesse Marcel
   (b) Karl Pflock
4. What major flaw did critics point out regarding all the witness accounts collected?
   (a) they came a minimum of thirty-one years after the event
   (b) they were contradicted by radar logs
5. According to the Air Force's 1997 report, what contributed to the reports of recovered alien bodies?
   (a) memories of military accidents and anthropomorphic test dummies
   (b) secret biological warfare experiments
6. During which holiday weekend does the City of Roswell host its annual Roswell UFO Festival?
   (a) July Fourth weekend
   (b) Memorial Day weekend

Story 9

1. From which empire was the tulip introduced to Europe in the mid-sixteenth century?
   (a) the Ottoman Empire
   (b) the Persian Empire
2. What was responsible for the vivid, multicolored patterns on the most sought-after tulip petals?
   (a) a tulip-specific virus
   (b) a rare soil mineral
3. How long does it take for a tulip seed to form a flowering bulb?
   (a) seven to twelve years
   (b) one to two years
4. Where did merchants and traders gather to buy and sell tulip futures contracts?
   (a) in taverns
   (b) in royal courtyards

5. Who wrote the 1841 book *Extraordinary Popular Delusions and the Madness of Crowds*?
   (a) Charles Mackay
   (b) Charles de l'Ecluse
6. Why did Dutch courts refuse to enforce payments on tulip contracts after the market crashed?
   (a) they regarded the debts as contracted through gambling
   (b) the government had outlawed flower sales

Story 10

1. In what person did Dr. Georges Gilles de la Tourette first describe the disorder?
   (a) an eighty-six-year-old French noblewoman
   (b) a ten-year-old English schoolboy
2. What condition do parents often mistake sniffing tics for before a formal diagnosis?
   (a) seasonal allergies
   (b) a chronic sinus infection
3. What is the term for complex vocal tics that involve uttering swear words?
   (a) coprolalia
   (b) echolalia
4. For at least how long must a patient display both motor and vocal tics to receive a formal diagnosis of Tourette's?
   (a) at least one year
   (b) at least six months
5. What is the main purpose of using neuroimaging studies like MRIs or CT scans during evaluation?
   (a) to rule out other conditions
   (b) to measure the severity of the tics
6. Females with a genetic predisposition for the disorder are more likely than males to exhibit which type of symptoms?
   (a) obsessive-compulsive symptoms
   (b) severe motor tics